

Language of Zelda: Facilitating Language Learning Practices Using ChatGPT

Sukran Karaosmanoglu
Human-Computer Interaction,
Universität Hamburg
Hamburg, Germany
sukran.karaosmanoglu@uni-
hamburg.de

David Kraus
Human-Computer Interaction,
Universität Hamburg
Hamburg, Germany
david.kraus@studium.uni-
hamburg.de

Elisabeth L Fittschen
Human-Computer Interaction,
Universität Hamburg
Hamburg, Germany
elisabeth.fittschen@studium.uni-
hamburg.de

Henrik Nickelmann
Human-Computer Interaction,
Universität Hamburg
Hamburg, Germany
henrik.nickelmann@studium.uni-
hamburg.de

Hande Eyicalis
Human-Computer Interaction,
Universität Hamburg
Hamburg, Germany
hande.eyicalis@studium.uni-
hamburg.de

Anna Tomko
Human-Computer Interaction,
Universität Hamburg
Hamburg, Germany
anna.tomko@studium.uni-
hamburg.de

Frank Steinicke
Human-Computer Interaction,
Universität Hamburg
Hamburg, Germany
frank.steinicke@uni-hamburg.de



Figure 1: Link (protagonist) and Navi (blue orb) in the *Language of Zelda* game world.

ABSTRACT

The Language of Zelda is an educational game that re-imagines “The Legend of Zelda: A Link to the Past” for French language learning. With the integration of ChatGPT for non-player characters (NPCs), the game allows players to interact with NPCs to practice French through gameplay, puzzles, and quests. Our approach bridges the gap between declarative and procedural language knowledge, offering an engaging, immersive learning experience. The game’s adaptive dialogues cater to various proficiency levels, enhancing both education and entertainment values. Our work illustrates the

potential of combining AI with game-based learning to create effective, enjoyable language education tools.

CCS CONCEPTS

• **Software and its engineering** → **Interactive games**; • **Human-centered computing** → **Natural language interfaces**; • **Computing methodologies** → **Natural language generation**; • **Applied computing** → **Interactive learning environments**.

KEYWORDS

games, natural language processing, chatGPT, language, learning

ACM Reference Format:

Sukran Karaosmanoglu, Elisabeth L Fittschen, Hande Eyicalis, David Kraus, Henrik Nickelmann, Anna Tomko, and Frank Steinicke. 2024. Language of Zelda: Facilitating Language Learning Practices Using ChatGPT. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3613905.3648107>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3648107>

1 MOTIVATION

Learning of foreign languages is a crucial requirement in our globalized world. Although engagement is a key element of language learning [10], traditional methods often struggle to engage learners effectively. While studying grammar rules and memorizing vocabulary are necessary for language acquisition, they can sometimes be daunting and uninspiring [12]. Such approaches can present a significant hurdle, as it does not fully prepare learners for real-world communication. While language learning can happen in different contexts, language exposure in a country, where the target language is spoken, offers a more natural and engaging way to acquire language skills [5, 21]. But, unfortunately, this approach is often not feasible for everyone (e.g., due to health, or financial reasons). Even for those who can travel, exposure to a new language environment can be intimidating or culturally challenging [5].

Games provide engaging experiences for users and have been used for education [7, 11, 17]. Especially for language learning, there are research-based [11] and popular gamified applications, such as Duolingo [2]. These applications typically include gamification elements (e.g., leaderboards) to make the learning experience fun. But, they often do not have gameplay or provide natural language interaction for users. Yet, (i) exposure to spoken language is important for language learning, (ii) having a gameplay can offer more immersive experiences, (iii) natural language interaction can increase learning opportunities, such as via practicing more words.

To address these challenges, we have implemented *The Language of Zelda* inspired by a retro game “The Legend of Zelda: A Link to the Past” [1]. This game offers a playful experience that mimics the benefits of living in a French-speaking environment, without the associated costs or anxieties. In the game, users practice French in an interactive space and actively use French to navigate the game world, solve puzzles, and interact with non-player characters (NPCs). This interaction takes place in the form of text conversation, and the player can input free text (by constructing sentences and using vocabulary) to communicate with NPCs. This active engagement mirrors real-life language use, where understanding and responding to spoken and written language is crucial. Our game overcomes current limitations of gamified language learning applications by offering to practice language without any restrictions (instead of multiple choice questions), providing an immersive gameplay, and enabling natural interaction with NPCs. Overall, we summarize our contributions as follows:

- *The Language of Zelda*, inspired by a famous retro game,
- a gameplay for language learning,
- integration of ChatGPT into our game to provide an accessible natural language interaction.

2 RELATED WORK

In language learning, there are two types of knowledge: declarative and procedural [24]. While declarative knowledge involves the conscious recollection of knowledge [6, 24], the procedural one is used subconsciously during task performance [9, 24]. For instance, a learner might know grammar rules (declarative) better than a native speaker, but the native speaker is more adept at applying them (procedural) [13]. Therefore, strengthening procedural knowledge plays a crucial role for second language learners. In classroom

settings, this is typically achieved through speaking exercises (e.g., role play) [16]. However, these activities typically constitute only a small part of the curriculum and are limited to the classroom environment. Unlike these activities, there are publicly accessible applications that support language learning, such as Duolingo [2]. This application emphasizes vocabulary memorization and text translation, thus focusing more on declarative knowledge acquisition. However, there has been a recent shift towards integrating Natural Language Processing (NLP) in these tools with the release of ChatGPT (Chat Generative Pre-trained Transformer) [19].

ChatGPT is an artificial intelligence (AI) chatbot developed by OpenAI [19]. It uses NLP (Generative Pre-trained Transformer architecture) to generate human-like answers based on the users’ inputs (i.e., prompts). While ChatGPT is a relatively new tool [19], using NLP in language learning is not new. The U.S. military developed a system (*Tactical Language and Culture Training Systems* [14, 15]), requiring users to communicate with locals (i.e., NPCs) to complete tasks. This system used advanced voice recognition software for its time and provided limited but dynamic actions; the interview results point to positive experience of the users and report that in real life, users were able to perform various communicative tasks with the locals [14]. A recent commercial example of NLP use in language learning applications is Duolingo’s interactive ChatGPT-supported [19] feature called Duolingo Max [3]. This version allows users to role-play common scenarios, such as buying coffee, in a chat environment with AI. Although it is still in a demo phase with regional limitations, it represents a significant step forward.

Digital games can support language learning by offering a playful way to practice language [8, 11]. Research indicates that students retain language information better and are more engaged when there are rewards involved [20]. Thus, most of the language apps employ gamification elements, such as rewards and badges [18]. However, to our knowledge, integration of a gameplay into language learning applications remains sparse. By creating a game for language learning, we aim to provide users with an enjoyable, immersive experience, and an accessible natural dynamic language interaction practice that can be used regardless of location. We believe that newer advancements in NLP could enhance the realism, dynamism, and effectiveness of such games in language learning by creating more natural and responsive interactions.

3 THE LANGUAGE OF ZELDA

The Language of Zelda was inspired by the retro game “The Legend of Zelda: A Link to the Past” [1]. This game has timeless appeal and an intricate narrative that provides a familiar yet engaging backdrop for language learning. We developed a mini video game to showcase the core mechanics and narrative of *The Language of Zelda*. This demo gives a glimpse into how players interact with the game world and progress through their language learning journey. We use ChatGPT in our game for communication with NPCs. This innovative feature introduces dynamic, realistic dialogues, vastly enhancing both the educational value and the immersion of the game. Finally, our game aims to promote French language learning and therefore includes elements of French culture and folklore (e.g., cooking theme, names of the characters, quiz on traditional French cuisine). With our game, we target non-French speaking people.

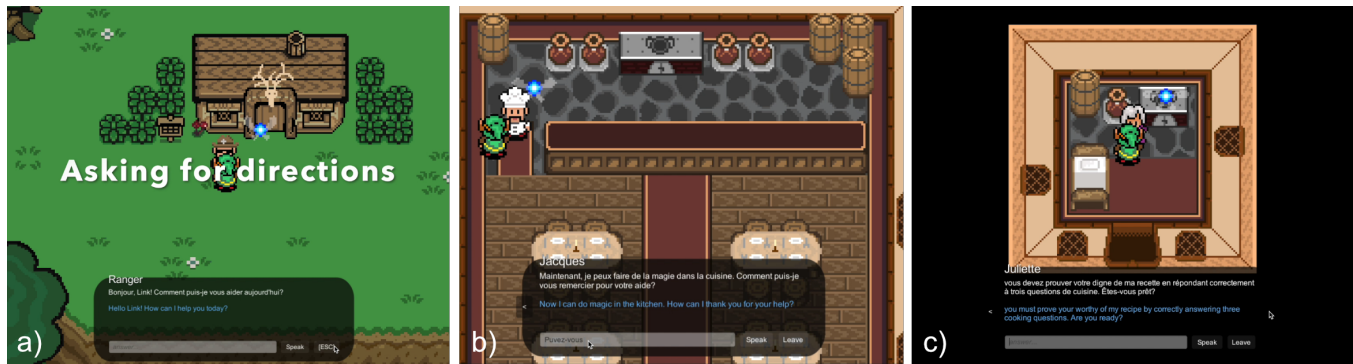


Figure 2: a) Link asks an NPC (ChatGPT integrated) for directions. b) Link talks with Jacques through text input that uses ChatGPT. c) Link talks with Juliette to secure a secret recipe. The text input window can be seen at the bottom of the images.

Gameplay. In our game, Link (protagonist) starts with amnesia and lands on a new, unfamiliar beach. This mirrors the player’s journey in learning a new language; just as Link is unfamiliar with his surroundings and history, the player is initially unfamiliar with the French language. The central quest revolves around Link’s mission to assist Jacques (see Figure 2b), a restaurant owner in a French-speaking village, by securing a secret recipe from Juliette (see Figure 2c), a renowned Mastercook. This quest integrates language learning into the game’s narrative, requiring players to engage in French conversations with NPCs to progress. To interact with NPCs, players use text input to type their questions and answers (Figure 2). Additionally, there are available tasks for players to complete as successfully answering a French cooking quiz.

A key feature of our game is the incorporation of Navi, a character represented as a blue orb that follows Link: Navi translates the dialogues from NPCs. When an NPC speaks, their dialogue appears in the standard format, and immediately below, Navi provides a translation (in English) in a distinct light blue font.

The game is viewed from a top-down perspective, allowing players to navigate the main character, Link, in multiple directions: up, down, left, right, and diagonally. This familiar style is instrumental in ensuring ease of navigation and enhancing the nostalgic experience for fans of the original series. Also, interactions within the game are straightforward and intuitive. The ‘C’ button is a multifunctional control used for various actions such as picking up objects or initiating conversations with NPCs. This simplicity in control ensures that the player’s primary focus remains on language learning and exploration, rather than on complex game mechanics.

The demo emphasizes player autonomy [22], allowing for various approaches to complete objectives, such as asking Juliette for help when solving the quiz, or being more lenient. However, all approaches involve writing in French. The interaction with NPCs through text input (supported via ChatGPT) is a critical element, immersing players in the French language and encouraging them to use their skills to overcome challenges and progress in the game.

Technical Specifications. The demo was implemented in Unity (v. 2021.3.19) [23] using a python backend for data handling/external API calls. The backend implements a “RESTful” API using flask. It accesses the OpenAI API and gpt-4-0613, gpt-4, and gpt-3.5-turbo

models, and the Google cloud API model translate_v2. The back-end handles all NPC and conversational data. First, the user input is checked for non-French input using fasttext. If the language inputted was not french with a confidence > 0.5 the user input is replaced with “something not understandable, certainly not french”. Then the NPC character description, currently relevant quest data, summarized conversation history, and current conversation are given to the gpt-4. The returned chat is then translated into English using the Google Translate API. After a conversation is finished, gpt-3.5-turbo generates a summary for future conversations, and gpt-4-0613 evaluates if any quests have been started or ended. Pre-defined “completion functions” in the quest description are called.

4 PLAYTESTING

After the development of *The Language of Zelda*, we conducted informal playtesting with a group (N=6) aged 14-24, most of whom had little to no prior knowledge of French. Players frequently utilized Google Translate to assist in generating text responses during gameplay, a practice that simulates real-world language use in a foreign environment. Our observations suggest a potential enhancement for the game: extending Navi’s translation feature. While we implemented a translation feature for NPCs’ answers, it may be useful to integrate translation or dictionary help for users’ answers, as they might not know or remember some French words. But, the lack of such features can also encourage players to think thoroughly and remember some words. Overall, the users’ feedback was overwhelmingly positive. All participants successfully completed the quest, indicating that the game’s design is effective in guiding learners, regardless of their initial language proficiency.

In a separate demonstration, we showcased a German version of the game (instead of French) to German-speaking graduate students. The aim of this demonstration was to gather feedback on the appeal of the game, translations, and dialogues. Our testing (approx. 4 people) confirmed that the game’s appeal extends beyond language learning; its interactive nature provides an enjoyable experience even for those already familiar with the language being taught.

5 CRITICAL REFLECTION

The Language of Zelda represents an innovative fusion of language learning with nostalgic gameplay. Our playtestings support that the

game appeals to different age groups. We believe that this interest could extend to older players. Moreover, unlike other commercial applications that only use game elements (e.g., Duolingo), our application goes one step further and features a game. This allows users to play a game while learning a language, and research supports the use of digital games for foreign language learning [8, 11].

Foreign language exposure plays an important role in language learning [5, 21]. Being exposed to a foreign language regardless of location provides accessibility to users. Our game offers the opportunity to practice French in front of computer screens. Our game is, of course, not comparable to visiting a foreign country and learning the language there, but it is an alternative that offers opportunities for players to practice. Since a new language environment can be stressful or culturally challenging for some people [5], we believe that such solutions can be beneficial. Moreover, in the long-term, such approaches can have economic and sustainability benefits.

Our game leverages ChatGPT for realistic and dynamic NPC dialogues, a novel feature in the realm of educational games; the potential of ChatGPT in educational games was also highlighted by [4]. This integration marks a significant step in the utilization of AI for personalized and game-based language learning. The game's use of ChatGPT for dialogue generation allows for a highly customized learning experience, as the dialogues can be tailored to each player's proficiency level and interests. This adaptive nature of the dialogues eliminates the need for pre-scripted interactions, enabling a more organic and engaging learning process.

Ethically, we believe that all users playing the game should be informed that their data is being processed by external parties. Moreover, it is important to inform players that they are interacting with AI-based NPCs so that they are aware that these characters are not controlled by humans, even if the interaction is natural. We also emphasize that ChatGPT may give unusual, incorrect, or meaningless responses, which users should be aware of [19].

We also note the limitations. The game currently focuses on French language learning, but expanding to include multiple languages is possible. This expansion, however, presents challenges in integrating cultural elements relevant to each language into the game aesthetics. Also, the game interaction is currently only limited to typing. However, incorporating speech recognition and synthesis is another promising avenue. Allowing players to practice spoken language and receive auditory feedback would create a more holistic language learning environment. Finally, our playtesting has a low sample size. Therefore, we recommend that future studies assess players' experiences with a higher sample size to understand the impacts of using ChatGPT-integrated educational games.

6 CONCLUSION

The Language of Zelda is an educational game that supports language learning. It has ChatGPT integration that enables dynamic, realistic interaction with NPCs and allows users to practice French. Our playtesting shows that players liked the game and the game has potential to support language learning. The game's potential for expansion and adaptation presents exciting opportunities for the future of educational gaming, promising to make language learning an engaging and enjoyable journey for a wide range of players.

ACKNOWLEDGMENTS

We would like to thank Silvina Kraus for her help with the game visuals and design. This work was supported by the European Union, the German Federal Ministry of Education and Research (BMBF), and the German Research Foundation (DFG).

REFERENCES

- [1] 1991. The Legend of Zelda: A Link to the Past. Game [SNES]. Release date: 21.11.1991.
- [2] 2012. Duolingo. Game [Mobile]. <https://www.duolingo.com> Accessed: 15.12.2023.
- [3] 2023. Duolingo Max. Game [Mobile]. <https://blog.duolingo.com/duolingo-max/> Accessed: 15.12.2023.
- [4] Ibrahim Adeshola and Adeola Praise Adepoju. 2023. The opportunities and challenges of ChatGPT in education. *Interactive Learning Environments* 0, 0 (2023), 1–14. <https://doi.org/10.1080/10494820.2023.2253858>
- [5] Heather Willis Allen and Carol Herron. 2003. A Mixed-Methodology Investigation of the Linguistic and Affective Outcomes of Summer Study Abroad. *Foreign Language Annals* 36, 3 (2003), 370–385. <https://doi.org/10.1111/j.1944-9720.2003.tb02120.x>
- [6] Niels Birnbaumer and Robert F. Schmidt. 2010. *Biologische Psychologie*. Springer, Berlin/Heidelberg/New York. https://doi.org/10.1007/978-3-540-95938-0_p_621
- [7] Marcus Carter, John Downs, Bjorn Nansen, Mitchell Harrop, and Martin Gibbs. 2014. Paradigms of games research in HCI: a review of 10 years of research at CHI. In *Proceedings of the First ACM SIGCHI Annual Symposium on Computer-Human Interaction in Play* (Toronto, Ontario, Canada) (CHI PLAY '14). Association for Computing Machinery, New York, NY, USA, 27–36. <https://doi.org/10.1145/2658537.2658708>
- [8] Stephan Franciosi. 2017. The Effect of Computer Game-Based Learning on FL Vocabulary Transferability. *Educational Technology & Society* 20 (01 2017), 123–133. <https://www.jstor.org/stable/jeductechsoci.20.1.123>
- [9] Michael Gazzaniga, Richard B. Ivry, and George R. Mangun. 2002. *Cognitive Neuroscience: The Biology of the Mind* (2 ed.). W.W. Norton & Company, Inc., New York.
- [10] Phil Hiver, Ali H. Al-Hoorie, Joseph P. Vitta, and Janice Wu. 2024. Engagement in language learning: A systematic review of 20 years of research methods and definitions. *Language Teaching Research* 28, 1 (2024), 201–230. <https://doi.org/10.1177/13621688211001289> arXiv:<https://doi.org/10.1177/13621688211001289>
- [11] Hsiu-Ting Hung, Jie Chi Yang, Gwo-Jen Hwang, Hui-Chun Chu, and Chun-Chieh Wang. 2018. A scoping review of research on digital game-based language learning. *Computers & Education* 126 (2018), 89–104. <https://doi.org/10.1016/j.compedu.2018.07.001>
- [12] Gladys Jean and Daphnée Simard. 2011. Grammar Teaching and Learning in L2: Necessary, but Boring? *Foreign Language Annals* 44, 3 (2011), 467–494. <https://doi.org/10.1111/j.1944-9720.2011.01143.x>
- [13] Keith Johnson. 1997. Language teaching and skill learning. Blackwell Oxford, England. <https://api.semanticscholar.org/CorpusID:60322212>
- [14] Lewis W. Johnson and Andre Valente. 2008. Tactical Language and Culture Training Systems: Using Artificial Intelligence to Teach Foreign Languages and Cultures. 1632–1639.
- [15] Lewis W. Johnson, Hannes Vilhjalmsson, and Prasan Samtani. 2005. The Tactical Language Training System. (01 2005).
- [16] Hayriye Kayi. 2006. Teaching speaking: Activities to promote speaking in a second language. *The internet TESL journal* 12, 11 (2006), 1–6.
- [17] Jeremy Lee, Kathleen Luchini, Benjamin Michael, Cathie Norris, and Elliot Soloway. 2004. More than just fun and games: assessing the value of educational video games in the classroom. In *CHI '04 Extended Abstracts on Human Factors in Computing Systems* (Vienna, Austria) (CHI EA '04). Association for Computing Machinery, New York, NY, USA, 1375–1378. <https://doi.org/10.1145/985921.986068>
- [18] Ana Manzano-León, Pablo Camacho-Lazarraga, Miguel A. Guerrero, Laura Guerrero-Puerta, José M. Aguilar-Parra, Rubén Trigueros, and Antonio Alias. 2021. Between Level Up and Game Over: A Systematic Literature Review of Gamification in Education. *Sustainability* 13, 4 (2021). <https://doi.org/10.3390/su13042247>
- [19] OpenAI. 2022. ChatGPT. <https://www.openai.com>. Accessed: 15.12.2023. Released: 30.11.2022.
- [20] Mark Peterson, Jeremy White, Maryam Mirzaei, and Qiao Wang. 2022. *A Review of Research on the Application of Digital Games in Foreign Language Education*. 1948–1971. <https://doi.org/10.4018/978-1-6684-3710-0.ch094>
- [21] Alex Pinar. 2016. Second Language Acquisition in a study abroad context: Findings and research directions. *Colombian Applied Linguistics Journal* 18 (12 2016), 83–94. <https://doi.org/10.14483/calj.v18n2.9204>
- [22] Richard M Ryan, C Scott Rigby, and Andrew Przybylski. 2006. The motivational pull of video games: A self-determination theory approach. *Motivation and emotion* 30, 4 (2006), 344–360. <https://doi.org/10.1007/s11031-006-9051-8> The IMI items are available online under this link: <http://selfdeterminationtheory.org>

- org/intrinsic-motivation-inventory/.
- [23] Unity Technologies. 2023. Unity Real-Time Development Platform. <https://unity.com/> LTS Release 2021.3.19f1..
- [24] Colin Thompson. 2019. Practice makes perfect? A review of second language teaching methods. *The Bulletin of the Graduate School of Josai International University* 22 (2019), 55–69.